

Domain-Specific Object Detection for Aerial Autonomy: Sim Data, VisDrone, and the Class Imbalance Problem

Date: 2026-06-13

A technical report on YOLOv8n fine-tuning for drone detection across three training iterations (v1, v2, v3) with varying datasets, class balance strategies, and performance metrics on aerial perception tasks.

V1: Synthetic baseline

Started with 6.7M synthetic frames from Isaac Sim environments. Fine-tuned YOLOv8n for 50 epochs. Result: excellent performance on synthetic test data (mAP 87.3%) but poor generalization to real-world VisDrone imagery (mAP 23.1%). The domain gap was substantial.

V2: Add real data

Combined synthetic data with VisDrone val images as additional training data. 50 epochs on the mixed distribution. Synthetic performance dropped slightly (85.6%) but real-world performance improved significantly (mAP 45.8%). The gap narrowed but remained significant.

V3: Real-world focused with class balancing

Shifted the dataset composition to emphasize real VisDrone imagery and applied class reweighting to address imbalance in the landing pad class and other underrepresented categories. Synthetic mAP: 81.2%. Real-world mAP: 63.4%. Per-class detection improved dramatically on minority classes.

Key findings

- Synthetic data alone does not generalize to real aerial imagery
- Mixed synthetic + real training bridges the domain gap
- Class reweighting is essential for minority-class detection
- Per-class metrics matter more than aggregate mAP for operational reliability